Raj Reddy

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Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, C++, TypeScript, Java, SQL

Software: Node.js, Docker, Podman, REST, AWS, Azure, Linux, Django, MongoDB, CI/CD, Microservices

Experience

HEXstream **Chicago, IL**
Software Engineering Intern *May 2022 – Aug 2022*

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
- Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

L3Harris Technologies **Greenville, TX**
Associate Software Engineer *June 2025 – Present*

- Built containerized microservices for high-frequency data ingestion, processing, and storage using Podman and serverless architecture on defense sensor streams.
- Developed signal processing algorithms in Python and C++ on embedded systems, meeting strict latency and resource constraints in collaboration with hardware teams.
- Automated CI/CD pipelines with static code analysis and integration testing, reducing deployment friction across engineering teams.

Doxy.me **Charleston, SC**
Software Engineering Intern *August 2024 – May 2025*

- Built transformer-based models with PyTorch and Llama, integrating WebRTC for real-time video streaming.
- Deployed inference pipelines on SageMaker & Fargate, improving diagnostic accuracy with reinforcement learning.
- Authored and maintained YAML-based configuration files for CI/CD pipelines, cloud infrastructure, and application deployments, enabling reproducible environments and streamlined releases.

University of Utah **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.